

Joseph Lang

heyjoelang@gmail.com

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PROFESSIONAL SUMMARY

Senior software engineer and technical lead with 12+ years shipping real-time 3D, XR, and AI systems. Architected Mursion's first on-demand AI simulation platform and led the Unity SDK at Leia Inc through an acquisition and a full architecture rewrite. Focused on turning ambiguous requirements into shipped, maintainable systems.

SKILLS

Systems & Real-Time: Python, WebRTC, LiveKit, Deepgram (Voice Activity Detection, STT), ElevenLabs TTS, JSON

3D Engines: Unity, C#, Unreal, C++, XR (VR, AR, MR), Mobile, Plugins, Rendering (URP, HDRP), Firmware Interfacing

Performance & Quality: Latency Optimization, Performance Profiling, Root Cause Analysis, Accessibility (WCAG 2.1)

Release Engineering: CI/CD (GitHub Actions, Jenkins, GameCI, Docker), Code Review, PR Workflow, CLI Tooling

Leadership: Cross-Functional Collaboration, Stakeholder Relations, End To End Ownership, Community Management

EXPERIENCE

Mursion, Senior Software Engineer

Oct 2024 - Sep 2026

- Architected Mursion's first **on-demand AI simulation platform** (24/7, now global) by integrating Deepgram speech-to-text, OpenAI reasoning, ElevenLabs voice synthesis, and real-time Unity 3D avatars in production
- Cut AI response **latency by 1.5 seconds**, tracing timestamped cross-service logs to a 1.2s Voice Activity Detection stall, then re-architecting the speech-to-text pipeline onto Deepgram Flux for integrated VAD
- Closed a WCAG 2.1 **keyboard-accessibility** gap by root-causing a WebRTC streaming security boundary silently dropping input events, then co-designing a vendor forwarding API and shipping the client-side integration
- Root-caused and resolved a production traffic-skew incident (**recognized by engineering leadership**) by isolating silent random misrouting between two builds during a Jenkins to ArgoCD migration
- Compressed report-to-root-cause time (**500+ reports triaged**) by building an in-simulation Jira pipeline auto-attaching session video, AI traces, and runtime diagnostics, handed off to QA as the owned workflow

Leia Inc, Enine Plugins Lead

Jul 2019 - Mar 2024

- Re-architected Leia's Unity SDK from **40 classes to 8 (-80% surface area)** with zero loss of core functionality, clearing technical debt, setting modular foundation for feature integration and cross platform integration
- Unified two device platforms after company acquisition by abstracting Windows and Android firmware APIs behind one C++/C# interface, shipping later features to both with no duplicated code
- Restructured render pipeline to support Unity URP, HDRP, and Post-Processing, built Unreal 5 plugin
- Led **3 Unity engineers** to a cleared bug backlog on projected timelines by standardizing releases end to end: Jira-to-branch traceability, PR review process, semantic versioning. Built a clear, systemic developer workflow

Gamehearts, Founder, Developer

Jul 2016 - Jul 2019

- Founded and ran a Unity XR consultancy shipping production AR/VR across automotive, healthcare, retail, film, and public-sector clients, owning technical scoping, delivery, and client relationships end to end
- Refactored Valeo Voyage XR within **5 weeks** by streamlining user inputs into one gesture, focusing on polish and high quality interactions and visuals. Project won Coyote Techno Innovation, **3rd place at CES 2019**
- Shipped Sevo The Dragon for Stanford Medicine Children's Health anesthesia stress reduction, owning programming, art sourcing, music integration, and timeline. Project featured in **NBC and Super Bowl ads**
- Built cinematic pipeline tooling for Sonder, one of the first animated films rendered in Unity, prototyping the pre-alpha Timeline system and feeding structured feedback that shaped development of a now core feature

- Built Inside Rodin's Hands, an AR exhibit at Stanford's Cantor Museum using image-based tracking to give visitors an X-ray view inside Rodin's sculptures. Featured on **Al Jazeera America**
- Shipped **4 interactive VR anatomy lectures** for Stanford Clinical Anatomy, converting static textbook content into spatial, explorable real-time 3D environments. Designed and developed end to end
- Drove clinical anatomy's shift from textbooks to interactive AR/VR by designing and building a real-time 3D anatomy viewer around professors' individual teaching workflows, replacing static slide decks
- Ported an iOS educational anatomy app to Windows, ZSpace, and Oculus, extending one content base across desktop, stereoscopic display, and VR runtimes without per-platform content rework

Awards

- **Techno Innovation 3rd Place:** Valeo Voyage XR, Coyote Automobile Awards, 2019
- **Best Use of IBM Watson:** Avatar Medic, IBM Watson AT&T Hackathon, 2018
- **Oculus Launchpad Winner:** NeuroExplorer VR, Meta, 2018
- **2nd Place:** Connect 4 Mixed Reality, Microsoft Holohacks, 2016
- **2nd Place:** Attack On Titan VR Showdown, High Fidelity Hackathon, 2016
- **1st Place:** Escape!, ZSpace App Challenge
- **Most Useful App:** Dino Labs, Google Project Tango App Challenge

EDUCATION
